UMEÅ UNIVERSITY



Lender and Borrower Interest Rates

Pricing European call option with asymmetric interest rates

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Abstract

This report studies the application of asymmetric interest rates on the pricing of an European call option. The differing interest rates represents the realistic setting where an investor generally pays a higher interest rate when borrowing than what the investor recieves as a lender. This is applied to the Black-Scholes model. We price the European call option using numerical methods based on a Partial Differential Equation framework and a Forward Backwards Stochastic Differential Equation framework. We conclude that the interest rate for lending money has no impact on the solution since we are in fact never lending money to finance the replicating trading strategy.

Glossary

BSDE Backwards Stochastic Differential Equation.

FBSDE Forward-Backwards Stochastic Differential Equation.

GBM Geometric Brownian Motion.

ITM In The Money.

OTM Out of The Money.

PDE Partial Differential Equation.

RMSE Root Mean Square Error.

SDE Stochastic Differential Equation.

Contents

1 Introduction **1**

1.1 Deterministic framework . . . 1
1.2 Probabilistic framework . . . 2
1.3 General parameters . . . 2

2 Theory **3**

2.1 Interpreting the PDE . . . 3
2.2 Interpreting the FBSDE . . . 5
2.2.1 The driver function . . . 5
2.2.2 BSDE and the hedging process . . . 6

3 Method **7**

3.1 PDE approach . . . 7
3.1.1 Finite Difference and Crank–Nicolson Scheme . . . 7
3.1.2 Determining interest rate . . . 8
3.1.3 Terminal condition . . . 9
3.1.4 Boundary conditions . . . 9
3.1.5 Approximating the option price . . . 10
3.1.6 Algorithmic summary for finding the option price . . . 10
3.2 FBSDE approach . . . 11
3.2.1 Simulating geometric Brownian motion . . . 11
3.2.2 Simulating backwards using BSDE . . . 11
3.2.3 Finding the hedging process . . . 12
3.2.4 Approximating the conditional expectation . . . 13
3.2.5 Algorithmic summary . . . 13

4 Result **15**

4.1 Deterministic framework . . . 15
4.1.1 Comparison to analytical solution . . . 15
4.1.2 Convergence of PDE solution . . . 16
4.2 Probabilistic framework . . . 17
4.2.1 Option price using the FBSDE approach . . . 17
4.2.2 Convergence of FBSDE solution . . . 18

5 Discussion **19**

5.1 Limitations . . . 19

6 Conclusion **20**

References **21**

1 Introduction

In this section we initially introduce the problem and the general framework. We proceed by introducing the approach specific methods.

This project studies option pricing in a modified Black-Scholes framework where the borrowing and lending interest rates differ. This reflects a more realistic financial setting, in which an investor typically receives a lower interest rate when lending capital than the rate paid when borrowing.

We consider a financial market consisting of a stock S_t and two bank accounts corresponding to lending and borrowing. The dynamics of these assets are given by

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t, \tag{1}$$

and

$$\begin{aligned} dL_t &= rL_t \, dt, \\ dB_t &= RB_t \, dt, \end{aligned}$$

where μ is the drift, $\sigma > 0$ is the volatility, and W_t is a standard Brownian motion. The constants r and R denote the lending and borrowing rates, respectively, with $R \geq r > 0$.

We focus on the pricing of a European call option with maturity $T > 0$ and strike price $K > 0$, with payoff

$$\Phi(S_T) = (S_T - K)^+.$$

In contrast to the classical Black-Scholes model, the presence of asymmetric interest rates introduces a nonlinearity in the pricing problem. In particular, the value of the replicating portfolio depends on whether the investor is in a borrowing or lending position, leading to a dependence of the dynamics on the sign of the cash position. As a result, the pricing problem can no longer be described by a linear model.

The objective of this project is to study this nonlinear pricing problem using two complementary approaches. First, we consider a deterministic approach based on Partial Differential Equation (PDE), which provides a grid-based numerical solution. Second, we formulate the problem in a probabilistic framework using Forward-Backwards Stochastic Differential Equation (FBSDE), allowing for a Monte Carlo-based solution.

1.1 Deterministic framework

For the PDE approach we use that the price of the option at time t is given by $Y_t = u(t, S_t)$ where u solves the following PDE:

$$\begin{cases} \partial_t u + \frac{1}{2}\sigma^2 s^2 \partial_{ss} u = \rho(u - s\partial_s u)\,(u - s\partial_s u), \\ u(T, s) = (s - K)^+, \end{cases} \tag{2}$$

where

$$\rho(x) = \begin{cases} r, & x \geq 0, \\ R, & x < 0. \end{cases} \tag{3}$$

In the model s is the stock price, σ is the volatility of the stock. K represents the strike price, T is the maturity time of the option. Finally r and R are the different interest rates described in Section 1. Where $S, \sigma, K, T, r, R \in \mathbb{R}_{>0}$.

The objective is to find the numerical solution $\tilde{u}(0, S_0)$. Which is the value of the contingent claim at time $t = 0$ and initial stock price $S = S_0$. One can clearly see that Equation (2) is similar to the standard Black-Scholes PDE but with r replaced by $\rho()$.

For further interpretation of the model see Section 2.1.

1.2 Probabilistic framework

For the FBSDE approach we introduce the following framework. Let

$$f(t, y, z) = -(yr - z\lambda + (y - \frac{z}{\sigma})^{-}(R - r)), \tag{4}$$

$$\Phi(x) = (x - K)^{+}. \tag{5}$$

Where $\lambda = \frac{\mu - r}{\sigma}$ and $x^{-} = \min(x, 0)$, $K, \mu, \sigma > 0$, $R \geq r > 0$. Then consider the following FBSDE

$$S_t = S_0 + \int_0^t \mu S_s ds + \int_0^t \sigma S_s dW_s, \quad S_0 = s_0 > 0, \tag{6}$$

$$Y_t = \Phi(S_T) + \int_t^T f(s, Y_s, Z_s) ds - \int_t^T Z_s dW s. \tag{7}$$

The components of the model is similar to Section 1.1, where S_t is the stock price process, Y_t is the option pricing process and Z_t is a hedging process. σ is the volatility of the stock, μ is the drift parameter of the stock process. K denotes the strike price of the option, T is the maturity time. R and r are the different interest rates.

The objective of this method is to find $\mathbb{E}(Y_0)$ representing the option price at time $t = 0$. For further details regarding the processes Y_t and Z_t as well as the function f and the Backwards Stochastic Differential Equation (BSDE) structure see Section 2.2.

1.3 General parameters

The problem we are solving uses various parameters. These parameters can be seen below in Table 1 where each row corresponds to a specific parameter.

Table 1: Parameters used for the implementations.

Parameter	Value
T	0.5
μ	0.06
σ	0.2
r	0.04
R	0.06
K	100
S_0	100

2 Theory

In this section we describe how the different frameworks introduced in Section 1 is interpreted and derived. Due to the numerical nature of this report some of the derivation details have been omitted. For further explanation see the mentioned references. Beginning in Section 2.1 the equations and components of the PDE approach is discussed, followed by Section 2.2 where the FBSDE method is interpreted.

2.1 Interpreting the PDE

The PDE from Equation (2) describes the dynamics of the option price given the asymmetric interest rates. The formulation of the PDE is identical to the formulation by Bergman (1995, pp. 483) where the author generalize the standard Black-Scholes model by introducing imperfect market conditions. An imperfect market condition is for instance differing interest rates. For the interpretation of the PDE the self-financing replicating strategy will be used. From Financial Mathematics it is known that the standard Black-Scholes model is complete. However due to the numerical focus of this report, and that the result of this section will not be used in the implementation, it is assumed that the Black-Scholes model with the non-linear term is also complete. This implies that there exists a self-financing replicating strategy for the contingent claim.

Consider Equation (2), it is clear that $u(t, s)$ represents the pricing process of the option. What might not seem obvious is what $u - s\partial_s u$ represents. From the description it can be inferred that it is the cash position since it regulates which interest rate is active in the equation. This will be explicitly derived.

Since the European call option is a contingent claim, consider the self-financing replicating strategy (ϕ, x) that replicates the option value. Let V_t be the value process of the replicating strategy, ϕ_t be the number of shares held and D_t be the money invested in the money market, or equavalently the cash position, then

$$Y_t = u(t, S_t), \tag{8}$$

$$V_t = D_t + \phi_t S_t. \tag{9}$$

Where the Expression (8) is the option price and Expression (9) is the self-financing repli-

cating strategy. Equation (1) describes the dynamics of S_t, thus applying Itô's formula on both equations yields

$$\begin{aligned} dY_t &= \partial_t u(t, S_t)dt + \partial_s u(t, S_t)dS_t + \frac{1}{2}\partial_{ss} u(t, S_t)d\langle S\rangle_t, \\ &= \partial_t u(t, S_t)dt + \partial_s u(t, S_t)(\mu S_t dt + \sigma S_t dW_t) + \frac{1}{2}\partial_{ss} u(t, S_t)\sigma^2 S_t^2 dt. \end{aligned}$$

Thus

$$dY_t = (\partial_t u(t, S_t) + \partial_s u(t, S_t)\mu S_t + \frac{1}{2}\partial_{ss} u(t, S_t)\sigma^2 S_t^2)dt + \sigma S_t \partial_s u(t, S_t)dW_t. \tag{10}$$

For finding the dynamics dV_t of the self-financing replicating strategy we follow the reasoning by Brigo et al. (2012, pp. 4-7). The authors show that for the trading strategy to be self-financing, the dynamics of D_t must capture the change in capital derived from selling and buying the stock. We omit further details but the dynamics of D_t will thus be

$$dD_t = -S_t d\phi_t - d\langle S, \phi\rangle_t + \rho(D_t)D_t dt. \tag{11}$$

Using Equation (11) and Itô's formula on the self-financing replicating strategy yields

$$\begin{aligned} dV_t &= \phi_t(\mu S_t dt + \sigma S_t dW_t) + S_t d\phi_t + \langle S, \phi\rangle_t + dD_t, \\ &= \phi_t(\mu S_t dt + \sigma S_t dW_t)\rho(D_t)D_t dt, \\ &= (\phi_t \mu S_t + \rho(D_t)D_t)dt + \phi_t \sigma S_t dW_t. \end{aligned}$$

Since the dynamics of the pricing process and the dynamics of the value of the self-financing replicating strategy must be equal, the process ϕ_t is found from solving the equality of the diffusion terms:

$$\sigma S_t \partial_s u(t, S_t)dW_t = \phi_t \sigma S_t dW_t. \tag{12}$$

Hence using the equivalence between the pricing process and the value of the replicating strategy results in

$$\partial_s u(t, S_t) = \phi_t. \tag{13}$$

Combining this result with Equation (8) and Equation (9), yields:

$$\begin{aligned} V_t &= \phi_t S_t + D_t \\ D_t &= V_t - \phi S_t \\ D_t &= u(t, S_t) - S_t \partial_s u(t, S_t) \end{aligned}$$

Thereby it has been shown that the cash position of the self-financing replicating strategy is the expression $u - s\partial_s u$ which coincides with the problem description. The rest of Equation (2) is the standard Black-Scholes model with an European call option.

2.2 Interpreting the FBSDE

Concerning the FBSDE approach, this subsection will explain all relevant components. Starting with Equation (5), this is clearly the European call option described in Section 1 and used in Section 1.1. Continuing with Equation (6), which is the same expression on integral form of Equation (1). This process S_t is the stock price, using Equation (7) it can be seen that Y_t is the option pricing process. The remaining components, namely, the Z_t process, the driver function f and the BSDE structure, might not be as easily identifiable.

2.2.1 The driver function

For interpreting the driver function f from Equation (4) it will be easier to start with the study the dynamics of the pricing process. Equation (7) can be differentiated backwards to derive the dynamics of Y_t, namely

$$dY_t = -f(t, Y_t, Z_t)dt + Z_t dW_t. \tag{14}$$

It can now see that the variable Z_t represents the diffusion part of the pricing process. We can also get an intuiton for the contribution of the driver function f.

Similarly to Section 2.1 consider the value process of the replicating strategy V_t. Using Equation (10) and the equivalence of the dynamics of the pricing process and the replicating strategy

$$Z_t = \sigma S_t \partial_s u = \sigma S_t \phi_t. \tag{15}$$

Thereby it is reasonable to infer that the Z_t process contains the riskiness of the replicating strategy since it equals the product of the volatility, stock price and the numbers of shares owned.

Equation (14) also motivates the rest of the components of the BSDE. Inserting the driver function f into Equation (14) yields

$$dY_t = (Y_t r - Z_t \frac{\mu - r}{\sigma} + (Y_t - \frac{Z_t}{\sigma})^-(R - r))dt + Z_t dW_t. \tag{16}$$

Consider the factor

$$Y_t - \frac{Z_t}{\sigma}.$$

Using Equation (15) and that $Y_t = u(t, S_t)$ from Section 2.1 gives

$$Y_t - \frac{Z_t}{\sigma} = u - \frac{S_t \sigma \partial_s u}{\sigma} = u - S_t \partial_s u.$$

Which is the same expression for the cash position that was derived in Section 2.1.

Using this result in the driver function f, provides intuition for the rest of the components. Lets consider only the drift component of the dynamics of Y_t in Equation (16). And consider the found transformations $u = Y_t$, $Z_t = \sigma S_t \partial_s u$ as well as $\partial_s u = \phi_t$ from Equation (13) being

the numbers of shares held. The expression for the drift becomes

$$
\begin{aligned}
dY_t &= (Y_t r + Z_t \frac{\mu - r}{\sigma} + (Y_t - \frac{Z_t}{\sigma})^{-}(R - r))dt + \dots, \\
dY_t &= \begin{cases} (Y_t r + Z_t \frac{\mu - r}{\sigma} + 0)dt + \dots & , \quad Y_t - \frac{Z_t}{\sigma} \geq 0, \\ (Y_t r + Z_t \frac{\mu - r}{\sigma} + Y_t R - Y_t r - R\frac{Z_t}{\sigma} + r\frac{Z_t}{\sigma})dt + \dots & , \quad Y_t - \frac{Z_t}{\sigma} < 0, \end{cases} \\
dY_t &= \begin{cases} (ur + \mu S_t \partial_s u - r S_t \partial_s u)dt + \dots & , \quad Y_t - \frac{Z_t}{\sigma} \geq 0, \\ (uR + \mu S_t \partial_s u - R S_t \partial_s u)dt + \dots & , \quad Y_t - \frac{Z_t}{\sigma} < 0, \end{cases} \\
dY_t &= \begin{cases} (\mu S_t \partial_s u + r(u - S_t \partial_s u))dt + \dots & , \quad Y_t - \frac{Z_t}{\sigma} \geq 0, \\ (\mu S_t \partial_s u + R(u - S_t \partial_s u))dt + \dots & , \quad Y_t - \frac{Z_t}{\sigma} < 0, \end{cases} \\
dY_t &= \begin{cases} (\mu S_t \phi_t + r(u - S_t \partial_s u))dt + \dots & , \quad Y_t - \frac{Z_t}{\sigma} \geq 0, \\ (\mu S_t \phi_t + R(u - S_t \partial_s u))dt + \dots & , \quad Y_t - \frac{Z_t}{\sigma} < 0. \end{cases}
\end{aligned}
$$

This means that the driver function f contributes to the pricing process Y_t the capital of the stock as well as the impact from borrowing or lending money.

2.2.2 BSDE and the hedging process

A reasonable question to ask is why the Z_t variable is needed for the BSDE. The reasoning for this follows from El Karoui et al. (1997, pp. 17-21). The full details will not be covered but the Z_t process is a result of the Martingale Representation Theorem. Where the authors construct a process representing the entire value of the contingent claim. Then showing that this is a martingale and a $\mathcal{L}^2$ process and thus applying the Martingale Representation Theorem. After reconstructing the option price process from this martingale they recover the expression for the BSDE seen in Equation (7). The proof contains further details concerning the existence and uniqueness of the two processes Y_t and Z_t due to Banach's fixed point theorem and Picard iterations.

In the context of this report we summarize the paper from El Karoui et al. (1997, pp. 18) as Z_t is the process ensuring that Y_t is a $\mathbb{F}$-adapted process such that $Y_T = \Phi(S_T)$.

Since it was shown that $Z_t = \phi_t \sigma S_t$, the Z_t process will be interpreted as the hedging strategy. To summarize the interpretation we can see that the two approaches use the same mechanics. Most notably the cash position and hedging strategy. Since the two models, appropriately model the same system, it is expected that these two approaches will provide the same result.

3 Method

In this section we describe how the problem was solved using the different methods. In Section 3.1 we describe the method used to solve the PDE framework. Then in Section 3.2 we discuss the methodology for the FBSDE approach.

3.1 PDE approach

In order to find a solution to Equation (2) we use a standard finite difference θ-scheme specifically the Crank-Nicolson numerical scheme with $\theta = \frac{1}{2}$. The sole irregular component is estimating $\rho(u - s\partial_s u)$. This will be done using a similar finite difference approach and can be seen in Section 3.1.2.

3.1.1 Finite Difference and Crank–Nicolson Scheme

The PDE is solved numerically by discretizing the time interval $[0, T]$ and the spatial domain $[S_{\min}, S_{\max}]$ using homogenous grids. N_s inner grid points is used for the spatial domain and N_t gridpoints for the time domain, such that

$$t_n = n\Delta t, \quad n = 0, \dots, N_t, \qquad s_i = i\Delta s, \quad i = 0, \dots, N_s + 1,$$

with

$$\Delta t = \frac{T}{N_t},$$

and

$$\Delta s = \frac{S_{\max} - S_{\min}}{N_s + 1}.$$

The numerical solution is denoted by $\tilde{u}_i^n \approx u(t_n, s_i)$. The iterative approach is found starting from the terminal time moving backwards to the initial time $t = 0$.

Derivatives are approximated using finite differences, with forward difference for the time domain, central difference for the spatial domain and central second order for the second order spatial derivative. The factors of the infinitesimal generator are identified from Equation (2) as

$$\bar{a} = \frac{1}{2}\sigma^2 s^2, \quad \bar{b} = \rho(u - s\partial_s u), \quad \bar{r} = -\rho(u - s\partial_s u).$$

Using the found factors and the finite differences, the discrete operator $A_{\Delta s}$ acting on $\tilde{u}_i^n$ is constructed.

Finally applying the θ-scheme and the matrix representation of the discrete operator the numerical scheme for the Crank-Nicolson scheme is found.

$$(I - \theta\frac{\Delta t}{(\Delta s)^2}A)\tilde{u}_{n-1} = (I + (1-\theta)\frac{\Delta t}{(\Delta s)^2}A)\tilde{u}_n. \tag{17}$$

Where I is the identity matrix, A is the matrix representation of the discrete operator $A_{\Delta s}$ and $\theta = 1/2$.

Since this scheme only covers the interior points for the spatial domain, auxiliary vectors are needed that affects the inner points with the boundary values.

This is done by constructing two $(N_s \times 1)$ vectors B_n and B_{n-1} containing the impact from the upper and lower boundaries on the finite differences.

$$B_n = \begin{pmatrix} \alpha_1 \tilde{u}_0^n \\ 0 \\ \vdots \\ 0 \\ \gamma_{N_s} \tilde{u}_{N_s+1}^n \end{pmatrix}, \quad B_{n-1} = \begin{pmatrix} \alpha_1 \tilde{u}_0^{n-1} \\ 0 \\ \vdots \\ 0 \\ \gamma_{N_s} \tilde{u}_{N_s+1}^{n-1} \end{pmatrix}. \tag{18}$$

Where

$$\alpha_i = \frac{\bar{a}_i}{(\Delta s)^2} - \frac{\bar{b}_i}{2\Delta s}, \quad \gamma_i = \frac{\bar{a}_i}{(\Delta s)^2} + \frac{\bar{b}_i}{2\Delta s}.$$

The vectors in Equation (18) are found by examining the finite differences for $i = 1$ and $i = N_s$ respecively.

Combining Equation (17) and Equation (18) with $\theta = 1/2$ yields the following final iterative scheme

$$(I - \frac{1}{2}\frac{\Delta t}{(\Delta s)^2}A)u_{n-1} = (I + \frac{1}{2}\frac{\Delta t}{(\Delta s)^2}A)u_n + \frac{\Delta t}{2}(B_{n-1} + B_n). \tag{19}$$

This is the central part of the numerical solution. In contrast to a standard Black-Scholes model with fixed interest rate we have a dynamic interest rate ρ which depends on both the spatial domain and the function value of u.

3.1.2 Determining interest rate

In order to solve Equation (3) and determine which interest rate to use in the numerical scheme we use a variation of the finite difference scheme discussed in Section 3.1.1.

Since

$$\rho(x) = \begin{cases} r, & x \geq 0, \\ R, & x < 0, \end{cases}$$

where $x = u - s\partial_s u$. Consider for $i \in \{1, \ldots, N_s\}$ the expression for the central difference of $\partial_s u_i$:

$$\begin{aligned} (u - s\partial_s u)_i &= \tilde{u}_i - s_i \frac{\tilde{u}_{i+1} - \tilde{u}_{i-1}}{2\Delta s} \\ &= s_i \frac{\tilde{u}_{i-1}}{2\Delta s} + \tilde{u}_i - s_i \frac{\tilde{u}_{i+1}}{2\Delta s} \end{aligned}$$

To extrapolate this for the entire spatial domain we represent the expression as the multiplication of a tridiagonal matrix with the column vector $\tilde{u}_n$. Such that

$$(u - s\partial_s u)(t_n) = A\tilde{u}_n + d_n.$$

Where the notation $(u - s\partial u)(t_n)$ represents $u - s\partial_s u$ evaluated at $t = t_n$. The vector d_n is a

$(N_s \times 1)$ auxiliary vector similar to B_i from Section 3.1.1 that impacts the inner points with the values at the boundaries. Such that

$$d_n = \begin{pmatrix} s_1 \frac{\tilde{u}_0^n}{2\Delta s} \\ 0 \\ \vdots \\ 0 \\ -s_{N_s} \frac{\tilde{u}_{N_s+1}^n}{2\Delta s} \end{pmatrix} \tag{20}$$

Since the value of $\rho()$ is needed to estimate the inner points of $\tilde{u}_{n-1}$, but $\tilde{u}_{n-1}$ is also needed for estimating $\rho()$, $\tilde{u}_{n-1}$ will be approximated by $\tilde{u}_n$.

3.1.3 Terminal condition

The terminal condition given by Equation (2) states at terminal time T the function value $u(T, S_T) = (S_T - K)^+$. This is rather simple to implement. Letting

$$\tilde{u}_i^{N_t} = \begin{cases} s_i - K, & s_i \geq K, \\ 0, & s_i < K, \end{cases} \tag{21}$$

satisfies the terminal condition.

3.1.4 Boundary conditions

The lower boundary condition is chosen simply as 0. Since if the spatial domain is chosen reasonably with regards to localization, then the lower bound will always be Out of The Money (OTM) meaning that that the option will not be exercised. This yields

$$\tilde{u}_0^n = 0, \, \forall n \in \{0, \ldots, N_t\}.$$

The upper boundary condition is problematic, with respect to the localization the upper bound will be chosen such that $S_{\max}$ is deeply In The Money (ITM) meaning that the option will always be exercised. For a standard Black-Scholes model this would mean that the value of the option is the difference between the current stock price and the discounted strike price. Namely $\tilde{u}_{N_s+1}^n = S_{max} - Ke^{-r(T-t_n)}$ for the discount rate r inherent to the model. In this case there are two different interest rates.

The larger borrowing interest rate R have been chosen for the upper boundary condition. Even though this might possibly overvalue the option price, if the spatial boundary is chosen adequately this will have a minimal effect on the solution. Hence the boundary conditions are

$$\tilde{u}_0^n = 0, \quad \tilde{u}_{N_s+1}^n = S_{max} - Ke^{-R(T-t_n)}. \tag{22}$$

3.1.5 Approximating the option price

Using the finite difference theta scheme produces an array containing the solution function to the PDE. The numerical function will have the same discretization as that of the time and space domains. Specifically the discretization of the space domain implies that the option price $Y_0 = \tilde{u}(0, S_0)$ corresponding to the initial stock price S_0 will surely not be included as a explicit point in the grid. In order to find the Y_0 corresponding to the spot price the second degree taylor polynomial will be used to approximate the value of $Y_0 = \tilde{u}(0, S_0)$.

The taylor polynomial of the second degree in continuous time for single variable is: [1]

$$P_2(x) = f(c) + f'(c)(x - c) + \frac{1}{2}f''(c)(x - c)^2. \tag{23}$$

Since the objective is to find the option price at $t = 0$, t is fixed at zero and Equation (23) is appropriately used. The finite differences are substituted for the continuous derivatives in Equation (23). The following approximation for the option price $Y_0 = \tilde{u}(0, S_0)$ is thus found as

$$u(0, S_0) \approx \tilde{u}_c^0 + \frac{\tilde{u}_{c+1}^0 - \tilde{u}_{c-1}^0}{2\Delta S}(S_0 - s_c) + \frac{1}{2}\frac{\tilde{u}_{c+1}^0 - 2\tilde{u}_c^0 + \tilde{u}_{c-1}^0}{(\Delta S)^2}(S_0 - s_c)^2. \tag{24}$$

Where c is chosen to be the index of s that is the closest to S_0. Specifically

$$c = \underset{0 \leq i \leq Ns+1}{\operatorname{argmin}} |s_i - S_0|.$$

3.1.6 Algorithmic summary for finding the option price

In order to clarify the implementation we will in this subsection describe the algorithmic implementation of the method for the PDE approach. The parameters used can be seen in Table 1 in Section 1.3.

The algorithm is initialized by choosing spatial boundaris S_{min} and S_{max} sufficently spaced from S_0 such that these values do not interfer with the inner points around the initial stock price S_0. This is proceeded by initializing the homogenously spaced time and space grids.

The function array $\tilde{u}$ is initialized as a $(N_s + 2 \times N_t + 1)$ array. The terminal values represented by the column vector $\tilde{u}_i^{N_t}$ are set according to Equation (21).

The iterative scheme starts with fixing the boundaries according to Equation (22) and then constructing the auxiliary vector d according to Equation (20). Using the boundaries multiplied with the auxiliary vector d we estimate ρ as the matrix product between the linear operator A_1 and the current column vector $\tilde{u}_n$ such that $\rho_{n-1} = A_1 u_n + d$. The scheme continues with finding the auxiliary vectors B_n and B_{n-1} described in Equation (18) and constructing the linear operator A_2 using the spatial domain, the boundaries and ρ_{n-1}. Lastly for the iteration, the linear system described by Equation (19) is solved.

This process is repeated for all $t_{N_t}, t_{N_t-1}, \ldots, t_2, t_1$ time steps until the column vector representing the $t = 0$ time is found. Finally Equation (24) is used to find the numerical approximation of $\tilde{u}(0, S_0)$.

[1] `https://en.wikipedia.org/wiki/Taylor%27s_theorem`. Accessed 2026-05-17.

The full implementation of the PDE method can be seen in the attached python file `PDE.py`.

3.2 FBSDE approach

As described in Section 1.2 we will use a FBSDE approach to solve the probabilistic framework for the pricing problem. This is done by initially simulating trajectories of the driving Black-Scholes model from Equation (1), then relying on the adaptedness of the pricing process, moving backwards in time to price the asset at time $t < T$.

3.2.1 Simulating geometric Brownian motion

The forward process is the standard Black-Scholes model seen in (6) which is a Geometric Brownian Motion (GBM). It is known that this Itô-process has the following explicit solution

$$S_t = S_0 e^{\int_0^t \mu - \frac{\sigma^2}{2} ds + \int_0^t \sigma dW_s}.$$

Beginning by discretizing time and introducing $\Delta t = t_{i+1} - t_i$ the following solutions for $t = t_{i+1}$ and $t = t_i$ are found

$$S_{t_i} = S_0 e^{\int_0^{t_i} \mu - \frac{\sigma^2}{2} ds + \int_0^{t_i} \sigma dW_s}, \tag{25}$$

$$S_{t_{i+1}} = S_0 e^{\int_0^{t_{i+1}} \mu - \frac{\sigma^2}{2} ds + \int_0^{t_{i+1}} \sigma dW_s}. \tag{26}$$

From (25) and (26) an exact iterative scheme is found for simulating trajectories

$$\begin{aligned} S_{t_{i+1}} &= S_0 e^{\int_0^{t_{i+1}} \mu - \frac{\sigma^2}{2} ds + \int_0^{t_{i+1}} \sigma dW_s} \\ &= S_0 e^{\int_0^{t_i} \mu - \frac{\sigma^2}{2} ds + \int_{t_i}^{t_{i+1}} \mu - \frac{\sigma^2}{2} ds + \int_0^{t_i} \sigma dW s + \int_{t_i}^{t_{i+1}} \sigma dW_s} \\ &= S_0 e^{\int_0^{t_i} \mu - \frac{\sigma^2}{2} ds + \int_0^{t_i} \sigma dW s + (\mu - \frac{\sigma^2}{2})(t_{i+1} - t_i) + \sigma(W_{t_{i+1}} - W_{t_i})} \\ &= S_{t_i} e^{(\mu - \frac{\sigma^2}{2})\Delta t + \sigma \Delta W_{t_i}} \end{aligned}$$

Using this we get an iterative approach for simulating the exact solution to the Black-Scholes process

$$S_{t_{i+1}} = S_{t_i} e^{(\mu - \frac{\sigma^2}{2})\Delta t + \sigma \Delta W_{t_i}}. \tag{27}$$

The implementation of this scheme relies of generating Brownian motion increments ΔW_{t_i} which will also be used in the BSDE.

3.2.2 Simulating backwards using BSDE

The goal for this subsection is to derive the iterative scheme for how the initial price Y_0 is estimated. Using Equation (7) and discretizing time such that $\Delta t = t_{n+1} - t_n$ using homogenous time grid the following equations are given

$$Y_{t_i} = \Phi(S_T) + \int_{t_i}^{T} f(s, Y_s, Z_s)ds - \int_{t_i}^{T} Z_s dWs,$$
$$Y_{t_{i+1}} = \Phi(S_T) + \int_{t_{i+1}}^{T} f(s, Y_s, Z_s)ds - \int_{t_{i+1}}^{T} Z_s dWs.$$

Using these and considering the difference between the time points yields

$$\begin{aligned} Y_{t_{i+1}} - Y_{t_i} &= \Phi(S_T) + \int_{t_{i+1}}^{T} f(s, Y_s, Z_s)ds - \int_{t_{i+1}}^{T} Z_s dWs, \\ &\quad - \Phi(S_T) + \int_{t_i}^{T} f(s, Y_s, Z_s)ds - \int_{t_i}^{T} Z_s dWs, \\ &= -\int_{t_i}^{t_{i+1}} f(s, Y_s, Z_s)ds + \int_{t_i}^{t_{i+1}} Z_s dW_s. \end{aligned}$$

Taking the conditional expectation on the difference with respect to $\mathcal{F}_{t_i}$ and applying zero-expectation property on the stochastic integral gives the following expression.

$$\mathbb{E}[Y_{t_{i+1}} - Y_{t_i}|\mathcal{F}_{t_i}] = \mathbb{E}[-\int_{t_i}^{t_{i+1}} f(s, Y_s, Z_s)ds + \int_{t_i}^{t_{i+1}} Z_s dW_s|\mathcal{F}_{t_i}],$$
$$Y_{t_i} = \mathbb{E}[Y_{t_{i+1}}|\mathcal{F}_{t_i}] + \mathbb{E}[\int_{t_i}^{t_{i+1}} f(s, Y_s, Z_s)ds|\mathcal{F}_{t_i}].$$

Proceeding by discretizing the lebesque integral and using that $f(t_i, Y_{t_i}, Z_{t_i})$ is $\mathcal{F}_{t_i}$-measurable, results in the following iterative scheme for Y_{t_i}:

$$\begin{aligned} Y_{t_i} &= \mathbb{E}[Y_{t_{i+1}}|\mathcal{F}_{t_i}] + \mathbb{E}[f(t_i, Y_{t_i}, Z_{t_i})\Delta t|\mathcal{F}_{t_i}], \\ &= \mathbb{E}[Y_{t_{i+1}}|\mathcal{F}_{t_i}] + f(t_i, Y_{t_i}, Z_{t_i})\Delta t. \end{aligned}$$

Since Y_{t_i} is needed to estimate Y_{t_i}, $Y_{t_{i+1}}$ will use as an approximation for the non-linear function. To complete the iterative scheme we need to find Z_{t_i} as well as $\mathbb{E}[Y_{t_{i+1}}|\mathcal{F}_{t_i}]$. The final iterative scheme for the FBSDE approach is then

$$Y_{t_i} = \mathbb{E}[Y_{t_{i+1}}|\mathcal{F}_{t_i}] + f(t_i, Y_{t_{i+1}}, Z_{t_i})\Delta t. \tag{28}$$

3.2.3 Finding the hedging process

Since the hedging process Z_{t_i} is iteratively needed, the same discretization as in Section 3.2.2 will be used to find Z_{t_i}. Consider the difference between the time steps again, then multiplying both sides with the corresponding Brownian motion increaments ΔW_{t_i} used in Section 3.2.1.

$$\begin{aligned} Y_{t_{i+1}} - Y_{t_i} &= -f(t_i, Y_{t_i}, Z_{t_i})\Delta t + Z_{t_i}\Delta W_{t_i}, \\ Y_{t_i}\Delta W_{t_i} &= Y_{t_{i+1}}\Delta W_{t_i} + f(t_i, Y_{t_i}, Z_{t_i})\Delta t\Delta W_{t_i} - Z_{t_i}(\Delta W_{t_i})^2. \end{aligned}$$

Taking the conditional expectation with respect to the given information at $\mathcal{F}_{t_i}$ yields:

$$\begin{aligned}\mathbb{E}[Y_{t_i}\Delta W_{t_i}|\mathcal{F}_{t_i}] &= \mathbb{E}[Y_{t_{i+1}}\Delta W_{t_i}|\mathcal{F}_{t_i}] + \mathbb{E}[f(t_i, Y_{t_i}, Z_{t_i})\Delta t\Delta W_{t_i}|\mathcal{F}_{t_i}] - \mathbb{E}[Z_{t_i}(\Delta W_{t_i})^2|\mathcal{F}_{t_i}],\\ 0 &= \mathbb{E}[Y_{t_{i+1}}\Delta W_{t_i}|\mathcal{F}_{t_i}] + 0 - Z_{t_i}\Delta t,\\ Z_{t_i} &= \frac{1}{\Delta t}\mathbb{E}[Y_{t_{i+1}}\Delta W_{t_i}|\mathcal{F}_{t_i}].\end{aligned}$$

Where Y_{t_i}, Z_{t_i} and Δt are measurable with respect to $\mathcal{F}_{t_i}$ while $\mathbb{E}[\Delta W_{t_i}|\mathcal{F}_{t_i}] = 0$ and $\mathbb{E}[(\Delta W_{t_i})^2|\mathcal{F}_{t_i}] = \Delta t$. This is a result of that the expectation of Brownian motion increments are zero and that the quadratic variation of a Brownian motion is t. Thus similary to Section 3.2.2 a method for approximating the conditional expectation is needed in order to find the process Z_t.

3.2.4 Approximating the conditional expectation

In order to get the iterative scheme for both Y_{t_i} and Z_{t_i} we need to approximate the conditional expectation $\mathbb{E}[\cdot|\mathcal{F}_{t_i}]$. This is done using a regression based method. From theory it is known that there exists a function v such that $\mathbb{E}[Y|R] = v(R)$. We want to find the function $v(R)$ that approximates $\mathbb{E}[Y|R]$ where Y is a random variable in L^2 and R is the current information. From theory it is also provided that v can be given by $v = \lambda^T b^n(R)$ where $b^n(R)$ is a function of the current information and λ solves the following ordinary least-square problem.

$$\lambda = \underset{\alpha\in\mathbb{R}^n}{\operatorname{argmin}}\,\mathbb{E}[|\alpha^T b^n(R) - Y|^2]$$

For this implementation the second degree polynomial is used to approximate the conditional expectation. The column vector $Y_{t_{i+1}}$ will be regressed on the current information S_{t_i} which is the current stock price. Since S_t is markovian S_{t_i} is the only variable neccesary. Then predicting the conditional expectation using

$$\mathbb{E}[Y_{t_{i+1}}|\mathcal{F}_{t_i}] \approx \lambda^T b^n(S_{t_i}).$$

Intuitively this means that since the pricing process is a realistic adapted stochastic process, it cannot rely on the known future values, even though the implementation moves backwards in time. However it may use that the distribution of the future pricing process is known. Thus by regressing the future pricing values on the current information, the implementation is able to approximate the future value using the current information.

This method will be used on both $\mathbb{E}[Y_{t_{i+1}}|\mathcal{F}_{t_i}]$ and $\mathbb{E}[Y_{t_{i+1}}\Delta W_{t_i}|\mathcal{F}_{t_i}]$ where the response variable is simply replaced.

3.2.5 Algorithmic summary

In this section we clarify the implementation for the FBSDE approach and how it is used to solve the problem. The parameters used throughout the implementation can be seen in Table 1.

The implementation starts of by simulating the GBM using the exact Scheme (27) derived in Section 3.2.1. We use sufficently large number of trajectories M. For the number of timesteps N we keep in mind that errors from the regression based approximation of the conditional expectation will affect the result.

The iterative approach is initialized by fixing the $Y_{t_N} = \Phi(S_T - K)$, then proceed starting at $t_{N-1} = T - \Delta t$. The conditional expectation of Y_{t_N} is found using the regression based approximation described in Section 3.2.4. The hedging process $Z_{t_{N-1}}$ using the same regression based approximation in accordance with Section 3.2.3. The pricing process $Y_{t_{N-1}}$ is finally found according to the Scheme (28).

When the iteration is done $Y_{t_{N-1}}$ and $Z_{t_{N-1}}$ have been computed.

The implementation proceeds by iterating all $t_n = t_{N-1}, t_{N-2}, \ldots, t_1, t_0$ until the pricing process for all M trajectories at $t = 0$ have been found. The final result from the implementation is computed by taking the mean to estimate $\mathbb{E}[Y_0]$ as the solution to the pricing problem.

The full implementation can be seen in the attached python file `FBSDE.py`.

4 Result

In this section we present the result from the implementation of the two approaches. Initially the result of the PDE implementation from Section 3.1.6 can be seen in Section 4.1. This is followed by the result from the FBSDE approach from Section 3.2.5 which can be seen in Section 4.2.

4.1 Deterministic framework

When implementing the numerical scheme described in Section 3.1.6 using the parameters described in Table 1 we get the following price for the European call option:

$$Y_0 = 7.15447.$$

In order to visualize the found numerical smooth function $\tilde{u}(t, s)$ there is a surface plot provided below in Figure 1. The plot visualizes the function on the time domain $t \in [0, 0.5]$ and spatial domain $S \in [1, 1000]$.

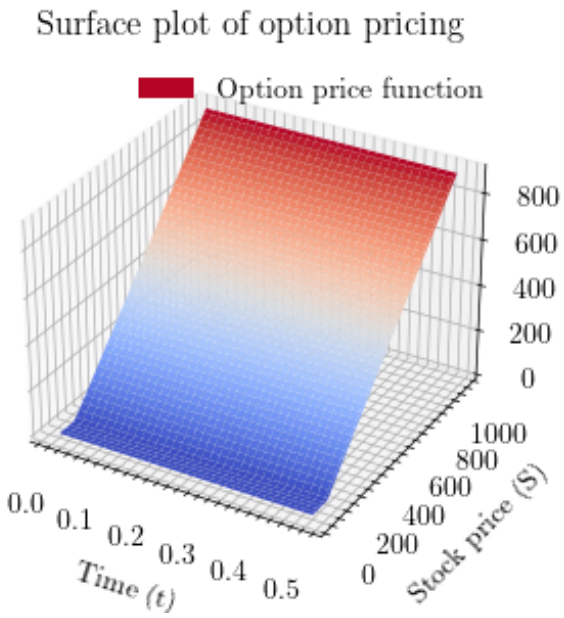


Figure 1: Surface plot for the pricing function $\tilde{u}(t, s)$.

As can be seen in Figure 1 the option is constantly zero when the option is OTM and positively linear when the option is ITM. In the spatial domain around the strike price we see that the option price starts growing from zero to the later linear growth.

4.1.1 Comparison to analytical solution

The resulting price $Y_0 = 7.15447$ is significant, since if we compare this to the analytical solution for the European call option in the standard Black-Scholes model with fixed interest

rate $r = R$ using the borrowing rate described as

$$
\begin{aligned}
Y_0 &= S_0\Phi(d_1) - Ke^{-RT}\Phi(d_2), \\
d_1 &= \frac{\log\frac{S_0}{K} + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}, \\
d_2 &= d_1 - \sigma\sqrt{T}.
\end{aligned}
$$

Where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution. With the parameters from Table 1 the analytical solution to the option pricing is

$$Y_0 = 7.15589$$

We will infer that the solution to our problem is indeed the analytical solution to the standard Black-Scholes model but with the borrowing rate only.

We proceed by comparing the solution curves of the numerical solution to the analytical solution. Below in Figure 2 there is a line plot describing the different methods at $t = 0$ over the spatial domain $S \in [1, 200]$.

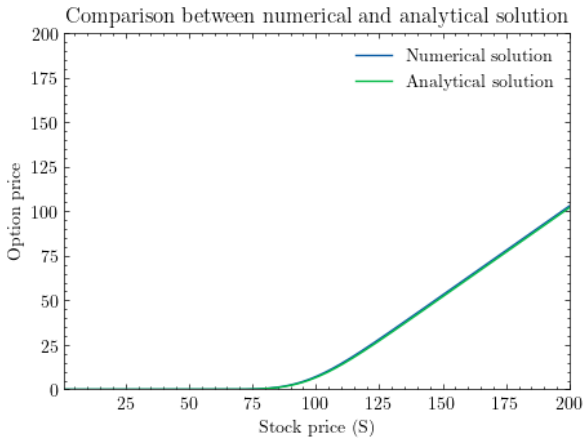


Figure 2: Line plot comparing the numerical solution to the analytical solution at $t = 0$.

As can be seen in Figure 2, the solution curves for the different methods align very closely for all points on the spatial domain. This provides further evidence that the analytical solution for only using the larger borrowing interest rate is the true solution to the problem.

4.1.2 Convergence of PDE solution

In order to further investigate the properties of the found solution we consider the Root Mean Square Error (RMSE) for comparing the numerical solution to the analytical solution. Below in Table 2 is the RMSE for increasing number of spatial grid points.

From Table 2 it can be seen that the error is decreasing as the number of grid points increases. To visualize this further there is a logarithm plot below in Figure 3 that shows the RMSE on the y-axis and the number of grid points on the x-axis.

Table 2: RMSE for increasing number of gridpoints

N_s	RMSE
20	0.42089
50	0.17893
80	0.07191
100	0.04792
200	0.00843
500	0.00065
800	0.00041
1000	0.00040

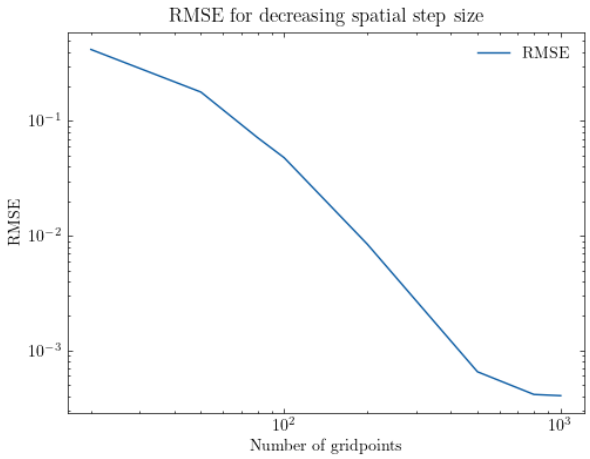


Figure 3: Logarithmic line plot over the RMSE for increasing number of gridpoints.

From Figure 3 one can clearly see that the RMSE converges as we increase the number of grid points or equivalently decrease the step size.

4.2 Probabilistic framework

In this section we show that the FBSDE method does indeed give the same solution as the PDE framework.

4.2.1 Option price using the FBSDE approach

Since the FBSDE approach gives an inherently stochastic result, each $\mathbb{E}[Y_0]$ will be treated as a random variable and we perform a plain Monte Carlo estimate with standard parameters $N = 100$ time steps, $M = 200000$ trajectories and $n = 100$ iterations.

Even though each FBSDE run is by itself a Monte Carlo trial using the M trajectories it is neccesary to perform multiple FBSDE runs. This is purely a result from using the regression based method where every trajectory affects every other. Thus the sample paths are not independent and independece is needed for the Monte Carlo estimate.

With these parameters the implementation yields the following estimate

$$Y_0 = 7.15568.$$

This option price and the variance of Monte Carlo estimate provides the following confidence interval using $\alpha = 0.05$

$$Y_0 \in [7.1520, 7.1593].$$

4.2.2 Convergence of FBSDE solution

In order to verify the convergence of the FBSDE approach we proceed similar to Section 4.1 and study the RMSE with increasing number of trajectories. In accordance with the previously found result, namely, that the true value of the option is the analytical solution using only the borrowing rate R. Thus the true value used for the RMSE is the analytical solution.

Below in Table 3 is the RMSE for increasing number of trajectores presented. Every instance is the Monte Carlo estimate using $N = 100$ time steps, $n = 100$ Monte Carlo iterations and varying number of trajectories M according to the table.

Table 3: RMSE for increasing number of trajectories

M	Y_0	RMSE
20 000	7.1615	0.0654
40 000	7.1544	0.0512
60 000	7.1623	0.0402
80 000	7.1593	0.0317
100 000	7.1498	0.0270
120 000	7.1567	0.0272
140 000	7.1579	0.0254
160 000	7.1565	0.0251
180 000	7.1566	0.0209
200 000	7.1592	0.0217

From Table 3 it can be seen that the RMSE decreases as we increase the number of trajectories for each FBSDE run. Below in Figure 4 the result from Tabel 2 have visualized. The diagram shows a logarithmic line plot with the RMSE on the y-axis and the number of trajectories on the x-axis. Where the number of trajectories ranges from 20000 to 200000.

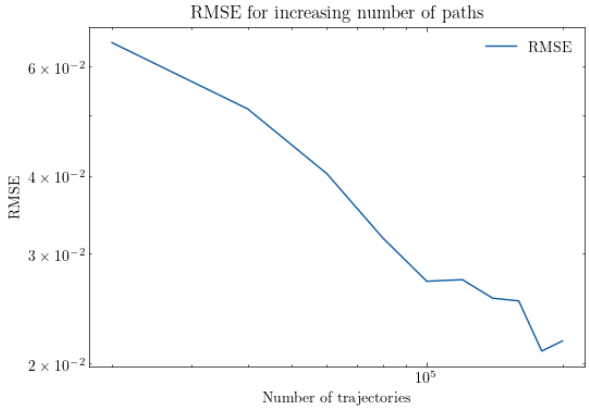


Figure 4: Logarithmic line plot of the RMSE for increasing number of trajectories.

From Figure 4 it can be seen that the RMSE decreases as the number of trajectories

increases.

Thus we conclude that the FBSDE numerical solution converges to the same solution as the PDE approach. Thus the implementations coincides with each other as well as the analytical solution for the Black-Scholes market model for a European call option with only the borrowing interest rate.

5 Discussion

In this section we discuss the finding of the report. Specifically the solution to the option pricing with asymmetric interest rates and the connection between the interpretation of the equations to the solution.

In Section 4 the result provided two notable conclusions. The first is that both implementations, the PDE and FBSDE, reached the same option price. Using the intepretation from Section 2, this is what was expected since it was shown that both frameworks model the same problem. Furthermore, given that the methods reached the same result further validates each model especially when comparing to the analytical solution.

The convergence tests performed in Section 4.1.2 and Section 4.2.2 provided ample support for the claim that the implementations both converge. And more explicitly that the numerical solutions converge to the analytical solution of the Black-Scholes model of a European call option. Even though there might have been a small discrepency error when comparing the PDE numerical solution to the analytical solution. This is probably a discretization error, both inherent to the numerical method, but also due to the approximation of the cash position.

The second notable conclusion is that the asymmetric interest rate does not affect the option price. This also implies that a more efficient model to solve the problem would be to remove the dynamics of the changing interest rate completely. The reasonable inference to why the lending rate has no effect on the option price must be that the replicating trading strategy never lends any money. It only borrows money to finance the trading strategy.

5.1 Limitations

Due to the limited time frame of the assignment there are some areas that were disregarded in favor of completing the assignment. Firstly, which is inherently theoretical, the impact of the drift parameter μ in Equation (1) should be analyzed in how that affects the solutions.

Secondly, some further experimentation regarding the convergence would be needed. For this report simulation sizes where limited by the capacity of the machine. This could have been remedied by using University provided cluster computing and further parallelization. With these adjustments further analysis regarding the convergence could have been done.

Even with these limitations this report still answers the problem stated in the assignment. Namely finding the numerical solution methods for the PDE and FBSDE approaches. As well as pricing the European call option and infering the explicit solution.

6 Conclusion

To summarize the report we have found that the lending rate does not affect the pricing of an European call option using the Black Scholes model with asymmetric interest rates. We derived two numerical approaches for pricing the contingent claim. The first based on PDE using the Black-Scholes model with a non-linear term depending on the cash position. The second approach based on Monte-Carlo methods and FBSDE specifically. Both implementations derived the same result, namely that the numerical solution coincides with the analytical solution for only using the borrowing interest rate.

We infered from the result that the lending rate did not affect the option price and we discussed the intuition behind always having to borrow money to finance the replicating strategy.

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